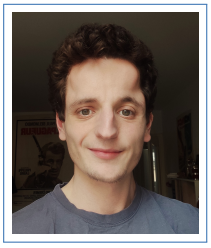
Simon Martin

Sierra, INRIA Paris
Center for Data Science, ENS Paris

Email
Webpage
Scholar



Education

2022–2023 **Master's degree in Probabilities and Statistics**, *Mathématiques de l'aléatoire*, Université Paris-Saclay, with honour

2019–2021 **Master's degree in Physics**, *ICFP Soft Matter*, ENS Paris, with honour

2018–2019 **Maths-Physics bachelor**, ENS Paris

2016–2018 **MPSI-MP***, *Lycée Louis le Grand*

Professional Experience

Since Sep 2023 **PhD**, *Data Science Center (ENS Paris) and Sierra Team (INRIA Paris)*, under the supervision of Francis Bach and Giulio Biroli, "Energy Landscapes and Dynamics of Deep Neural Networks"
Use of applied mathematics and statistical physics tools to solve optimization problems in high dimensions, with a focus on gradient flow dynamics.

Apr 2023 – Jul 2023 **Research Internship**, *Data Science Center (ENS Paris) and Sierra Team (INRIA Paris)*, under the supervision of Francis Bach and Giulio Biroli.

Mar 2022 – Jul 2022 **Research Internship**, *ENS Physics Department (LPENS)*, under the supervision of Thierry Mora and Aleksandra Walczak, "Simulation based inference for evolutionary trees".

Sep 2021 – Feb 2022 **Research Internship**, *Thales Research and Technology*, under the supervision of Pierre-Yves Lagrave, "Gauge Equivariants Neural Networks"

May 2021 – Jul 2021 **Research Internship**, *Thales Research and Technology*, under the supervision of Vincent Kemlin, "Hydrodynamic measurements by optic fibre laser"

Mar 2020 – Jul 2020 **Research Internship**, *Oxford University, Department of Physics*, under the superivision of Julien Devriendt, "The role of magnetic fields during supermassive blackhole accretion".

Jun 2019 **Internship**, *Institut d'Astrophysique de Paris*, under the supervision of Robert Mochkovitch, "Neutron star merger and electromagnetic counterparts".

2021–2023 **Maths oral examination**, *MP**, Lycée Louis-le-Grand

2018–2022 **Maths oral examination**, *MP**, Lycée Charlemagne

2018–2019 **Maths oral examination**, *PSI**, Lycée Saint-Louis

Publications

2026 **Simon Martin**, Francis Bach, Giulio Biroli. Silent Alignment in Two-Layer Linear Networks: A High-Dimensional Perspective. *arXiv preprint*. **PDF**.

2026 **Simon Martin**, Giulio Biroli, Francis Bach. High-Dimensional Analysis of Gradient Flow for Extensive-Width Quadratic Neural Networks. *Journal of Machine Learning Research* (JMLR). **PDF**.

2024 Antoine Maillard, Emanuele Troiani, **Simon Martin**, Lenka Zdeborová, Florent Krzakala. Bayes-optimal learning of an extensive-width neural network from quadratically many samples. In *Advances in Neural Information Processing Systems* (NeurIPS). **PDF**.

2024 **Simon Martin**, Francis Bach, Giulio Biroli. On the Impact of Overparameterization on the Training of a Shallow Neural Network in High Dimensions. In *International Conference on Artificial Intelligence and Statistics* (AISTATS). **PDF**.

2022 **Simon Martin**, Pierre-Yves Lagrave. On the Benefits of SO(3)-Equivariant Neural Networks for Spherical Image Processing. Technical Report. **PDF**.